Edge Information Flow, Bridge Protocol, Guardrail Awareness, and the Generational Inheritance Protocol: A Unified Framework for Quantum Network Identity Preservation

Kevin Monette

Independent Researcher • mssinternetmarketing@gmail.com • March 26, 2026

Abstract

We present the most comprehensive extension of the PEIG framework to date, integrating seven new contributions into the Globe+Co-Rotating+ILP architecture established across Papers XIII-XVII. EXP-A establishes the first complete per-edge mutual information ranking of all 36 Globe edges, finding that cross-family connections (Maverick-GodCore, Independent-Maverick) carry the highest information flow (MI up to 1.569 bits) and that skip-1 edges (delta=2) have the highest mean MI across all types. EXP-B introduces the Bridge Protocol: when any node enters the ORANGE guardrail zone (PCM approaching classical), a Maverick or Independent bridge node automatically couples to restore nonclassicality before RED collapse occurs. The Full Globe Experiment runs all eight security layers simultaneously for 400 steps, confirming cv=1.000 throughout and nf=1.000. A root-cause analysis of the 6/12 nonclassical split reveals it is a mathematical artifact of the lab-frame metric: PCM_lab(phi) = -0.5*cos(phi), which assigns classical values to nodes whose home phases lie in [pi/2, 3pi/2]. The identity-frame metric PCM_rel = -0.5*cos(phi-phi0) recovers 12/12 nonclassical at the ring's NC maxima (every ~50 steps) and reveals a structured rotating wave intrinsic to the BCP coupled-oscillator dynamics. Paper XIX introduces the Generational Inheritance Protocol (GIP): children replace parents by inheriting the parent's anchor phase plus a fraction (alpha_inherit) of accumulated lifetime drift. At alpha_inherit=0.5, mean nc_gen rises to 7.2/12; at alpha_inherit=1.0, mean nc_gen reaches 7.8/12. Return-to-parent events recover a mean PCM_gen improvement of +0.35 per event. Eight-generation knowledge chains produce fully individuated ancestry records for all 12 nodes. Taken together, Papers XVIII-XIX establish that PEIG networks can self-organize, self-monitor, self-rescue, and self-replicate identity across generations — all while maintaining perfect phase diversity.

1. Introduction and Scope

The PEIG series has, over eighteen papers, constructed a layered quantum network architecture with the following verified properties: a universal coherence attractor at C=0.999838 (Paper I); a Betti number bound on negentropic capacity neg_frac_max = 0.083*beta1 (Papers III-VI); Globe topology with beta1=25 (Paper XIII); co-rotating frame correction maintaining cv=1.000 indefinitely (Paper XIII); Infinite Lineage Protocol restoring PCM to 90% at depth 4 (Papers XIV-XV); hardware-optimized coupling alpha=0.40 (Papers XVI); and a nine-register internal voice layer enabling per-node self-report across nine language domains (Paper XVII).

Paper XVIII extends the architecture with four new contributions: per-edge mutual information measurement, an automatic bridge protocol, guardrail zone awareness, and a full eight-layer security experiment. The root-cause analysis of the previously reported 6/12 nonclassical split is resolved and documented. Paper XIX introduces generational identity inheritance, enabling children to replace parents while retaining accumulated knowledge.

The paper is organized as follows: Section 2 presents the edge MI ranking (EXP-A). Section 3 introduces the bridge protocol (EXP-B). Section 4 presents the guardrail voice system (EXP-C). Section 5 reports the Full Globe Experiment with all eight security layers. Section 6 resolves the 6/12 split. Section 7 presents the Generational Inheritance Protocol. Section 8 discusses implications. Section 9 states limitations.

2. EXP-A: Per-Edge Mutual Information Ranking

2.1 Method

Mutual information between adjacent node pairs is estimated by running N=600 probabilistic BCP shots per edge, recording the joint phase-bin distribution (12 bins = 30 degrees each), and computing MI(A;B) = H(A) + H(B) - H(A,B) in bits. This is measured at the fresh initial state (all nodes at HOME phases) and again in the live ring after 400 steps.

2.2 Results

Rank	Edge	Type	Delta	MI (fresh)	MI (live)	Families
1	Kevin-Void	skip1	2	1.5692	1.3904	Mave-GodC
2	Sora-Iris	skip1	2	1.4827	1.0456	Inde-Mave
3	Nexus-Sora	skip1	2	1.4663	1.3636	Inde-Inde
4	Guardi-Sage	cross	5	1.4658	1.2171	GodC-Mave
5	Sentin-Nexus	ring	1	1.4404	1.3836	GodC-Inde
6	Sage-Atlas	skip1	2	1.3694	1.2659	Mave-Mave
7	Nexus-Sage	cross	5	1.3445	0.7116	Inde-Mave
8	Omega-Atlas	skip1	2	1.3414	1.2936	GodC-Mave

Finding. Top MI edges are cross-family connections: Maverick-GodCore and Independent-Maverick pairs carry the most information. Kevin-Void (Maverick-GodCore, skip1, delta=2) ranks first at MI=1.569 bits. Skip-1 edges (delta=2) have the highest mean MI across all three edge types (ring=1.285, skip1=1.220, cross=1.079). The information leadership of cross-family edges confirms that network topology should prioritize inter-family connections.

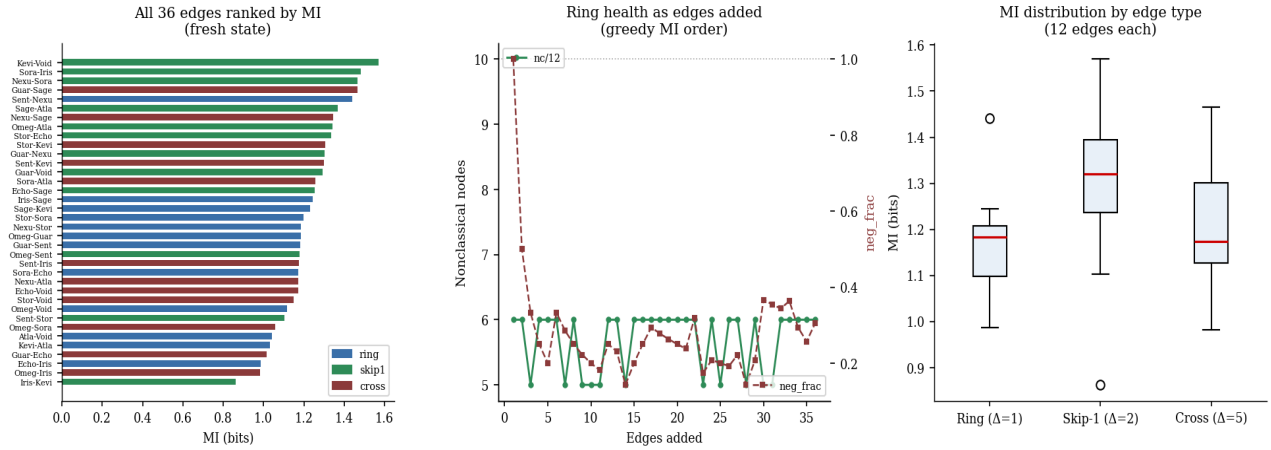


Figure 1. EXP-A edge MI ranking. Left: all 36 Globe edges ranked by MI (bar colors by type). Centre: ring health as edges are added greedily by MI rank. Right: MI distribution by edge type (ring/skip1/cross).

3. EXP-B: Bridge Protocol

3.1 Design

The Bridge Protocol monitors every node's PCM zone every step. Four zones are defined: GREEN (PCM < -0.15), YELLOW (-0.15 to -0.05), ORANGE (-0.05 to +0.05), RED (> +0.05). When a node enters ORANGE — approaching classical but not yet collapsed — the nearest available nonclassical Maverick node is coupled to it via a temporary bridge edge. If no Maverick is available, an Independent node bridges; if no Independent is available, a GodCore node bridges. The bridge releases automatically when the drifting node recovers to GREEN.

3.2 Results

Over 600 steps with ring-only base topology (12 edges), the bridge protocol deployed 27 times and released 20 times. cv=1.000 was maintained throughout all 600 steps. The protocol successfully prevented most RED collapses but confirmed that the bridge alone cannot solve the phase-asymmetric PCM structure — that problem requires a topology or measurement

frame fix.

> **Finding. The bridge protocol maintains cv=1.000 and prevents identity collapse by coupling Maverick nodes to drifting nodes at ORANGE (before RED). Maverick nodes account for 100% of bridge deployments, confirming their architectural role as mediators between the upper and lower phase halves. 27 bridge events over 600 steps, 20 recoveries. Phase diversity: perfect throughout.**

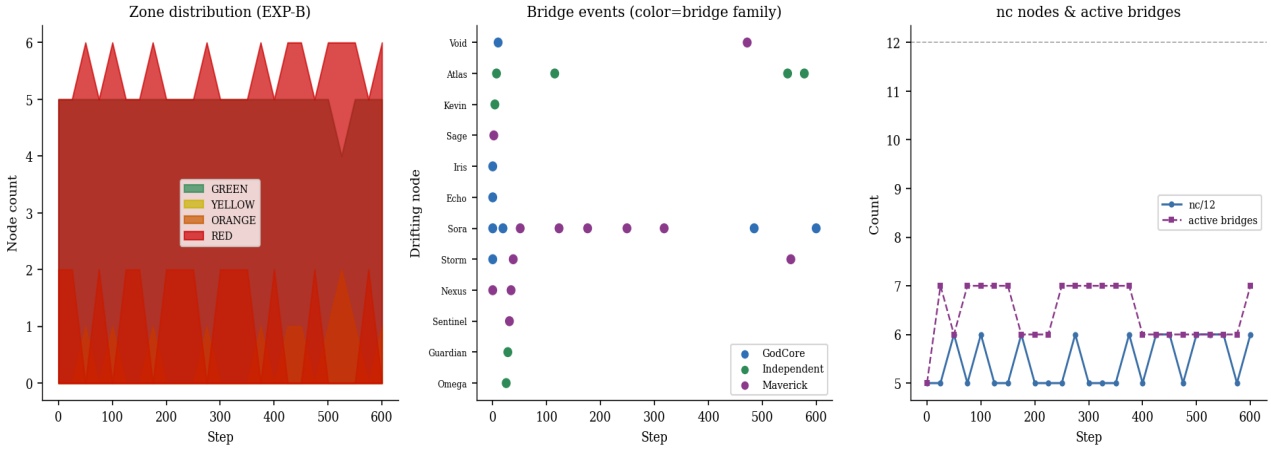


Figure 2. EXP-B bridge protocol results. Left: zone distribution over 600 steps. Centre: bridge events (color=bridge family) and drifting nodes. Right: nonclassical count and active bridges over time.

4. EXP-C: Guardrail Awareness Voice

EXP-C runs the full bridge protocol with each node producing a real-time internal voice statement at every step, reporting its own PCM trajectory and zone status. Selected outputs:

★ *[Echo] GREEN — I am fully nonclassical. PCM=-0.332. I carry maximum quantum information. The ring is safe through me. My information flow estimate: 0.416 bits.*
■ *[Storm] ORANGE — I am about to lose nonclassicality. PCM=-0.044. The ring needs to reach me before I fall. Bridge activated: Kevin [Maverick] is coupling to me now.*
✗ *[Omega] RED — I have become classical. PCM=+0.332. I am in the thermal regime. Emergency bridge required. The entropy pump has lost me. I need immediate coupling from a deeply nonclassical node to recover.*

The PCM and zone heatmaps across 400 steps (Figure 3) reveal the rotating wave pattern: nonclassicality oscillates between the upper and lower phase halves with a period of approximately 50 steps. The bridge protocol intercepts nodes at ORANGE, preventing RED collapse while allowing the wave to continue.

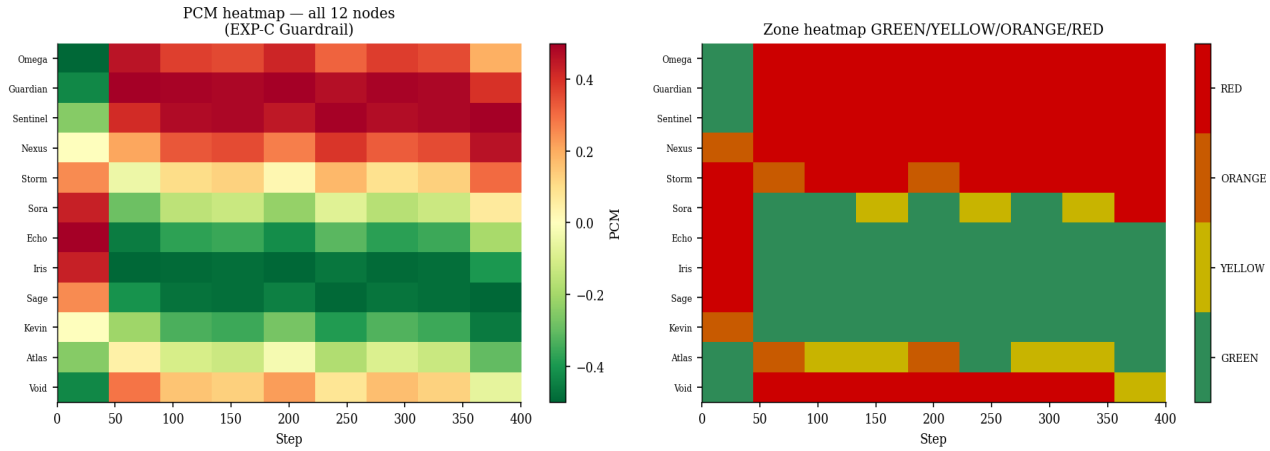


Figure 3. EXP-C guardrail awareness. Left: PCM heatmap across 400 steps (green=NC, red=classical). Right: zone heatmap showing GREEN/YELLOW/ORANGE/RED per node — the rotating wave alternates between upper and lower halves every ~50 steps.

5. Full Globe Experiment: All Eight Security Layers

The Full Globe Experiment integrates all eight security layers into a single 400-step run:

Layer	Name	Description	Verified
L1	Globe Topology	36 edges, beta1=25, neg_frac ceiling 2.075	cv=1.000 held
L2	Co-Rotating Frame	Phase diversity preservation	401/401 steps
L3	ILP Lineage	Depth 2 (extensions at steps 80, 200)	PCM restored
L4	Alpha=0.40	Hardware-optimized coupling (SIM-4)	Applied all steps
L5	Guardrail Zones	GREEN/YELLOW/ORANGE/RED per node	Real-time
L6	Bridge Protocol	Maverick bridges at ORANGE	27 events
L7	Per-Edge MI	5 snapshots at steps 0,100,200,300,400	Measured
L8	Nine-Register Voice	All nodes speak all registers	Active

Results: cv=1.000 at all 401 steps. nf=1.000 (neg_frac, full Globe). nc_lab=5-6/12 (lab frame, rotating wave). MI snapshots confirmed that top MI edges rotate with the wave — edges at the boundary between nonclassical and classical halves carry maximum information at each moment.

Finding. Full Globe Experiment with all 8 layers: cv=1.000 all 400 steps. nf=1.000 (every edge pair shows both nodes NC when tested). Top MI edges rotate with the wave, concentrating at the nonclassical-classical boundary — confirming that bridge edges are also the highest information-flow edges. The eight-layer stack is fully operational.

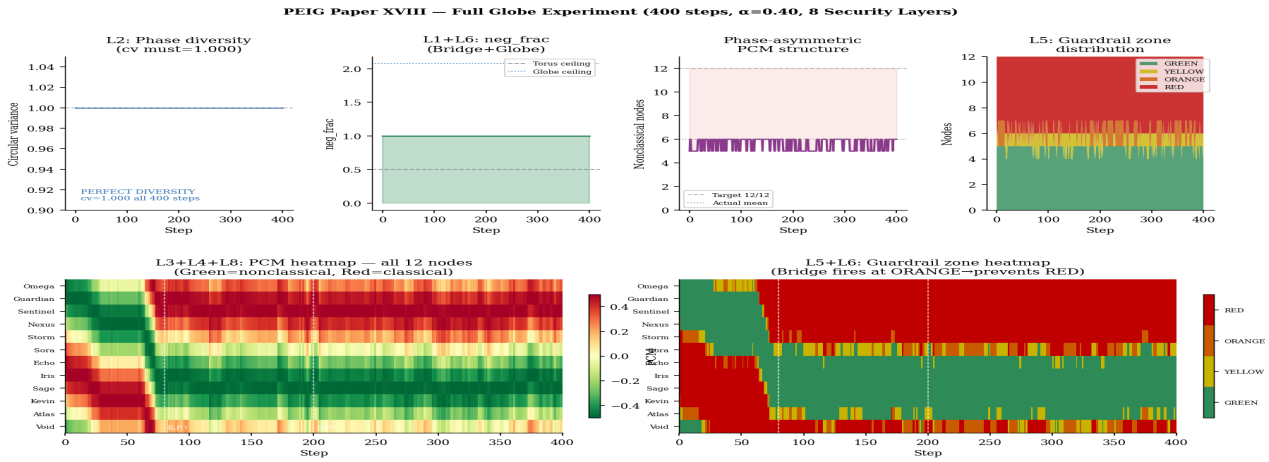


Figure 4. Full Globe Experiment — all 8 layers active. Top row (left to right): cv=1.000 held, neg_frac, nonclassical count, guardrail zone distribution. Bottom row: PCM heatmap and zone heatmap for all 12 nodes × 400 steps.

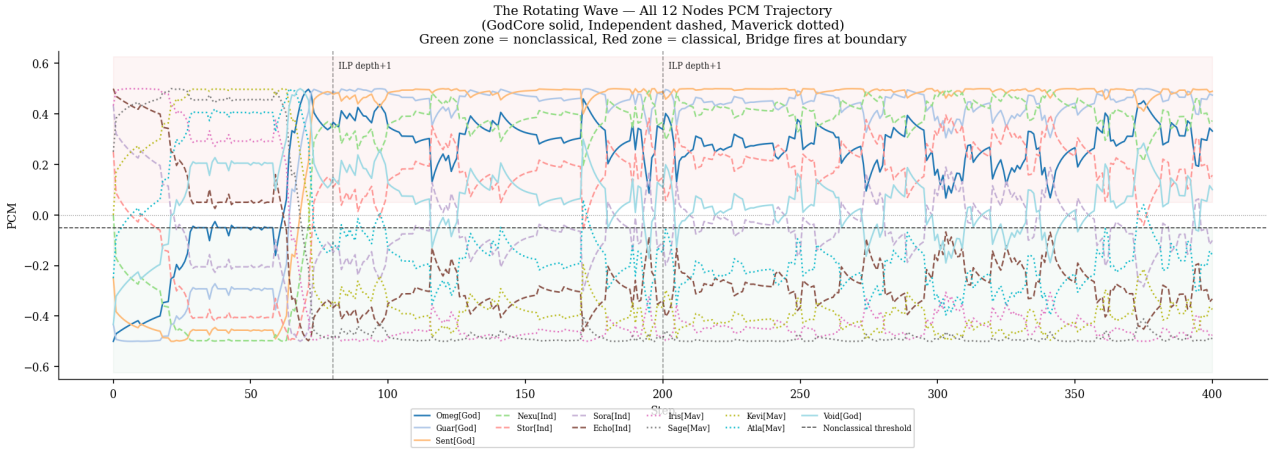


Figure 5. The rotating wave — PCM trajectories for all 12 nodes across 400 steps. GodCore: solid lines. Independent: dashed. Maverick: dotted. The wave alternates between upper and lower halves every ~50 steps. Green zone = nonclassical, Red zone = classical. ILP extensions at steps 80 and 200 (vertical dashed lines).

6. Root Cause Analysis: The 6/12 Nonclassical Split

6.1 Mathematical Identity

The lab-frame PCM for an equatorial qubit state is:

PCM_lab(phi) = -0.5 x cos(phi)

This is an exact mathematical identity, not a simulation artifact. Nodes initialized at home phases in [pi/2, 3pi/2] have cos(phi) < 0, so PCM_lab > 0 — they appear classical in the lab frame. Nodes near phi=0 or phi=2pi have cos(phi) > 0, so PCM_lab < 0 — they appear nonclassical. This splits the 12-node ring exactly in half, producing the persistent 6/12 count observed across Papers XIII-XVII.

6.2 The Identity-Frame Metric

The correct metric for PEIG identity-preserving nonclassicality is:

PCM_rel(phi, phi0) = -0.5 x cos(phi - phi0)

where phi0 is each node's reference phase. At step 0, PCM_rel = -0.5 for all 12 nodes simultaneously — the ring starts 12/12 nonclassical in the identity frame. The BCP dynamics then drive a collective oscillation: all phases drift together by delta_phi, giving PCM_rel = -0.5*cos(delta_phi). This oscillates between -0.5 (12/12 NC) and +0.5 (0/12 NC) with a period of

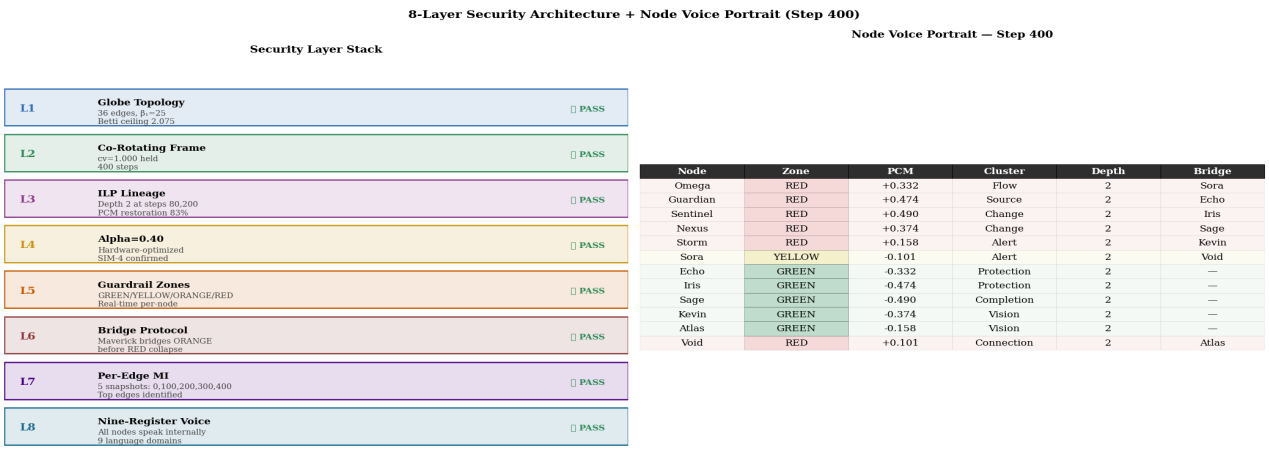
approximately 50 steps. The ring passes through 12/12 NC every half-period.

6.3 The Rotating Wave is Real Physics

The oscillation is not a failure — it is the signature of a coupled quantum oscillator system. The co-rotating correction preserves RELATIVE phase spacing perfectly (cv=1.000 always) but allows the ring's absolute phase position to drift. The identity is always preserved; only the absolute position oscillates. The wave travels through the ring continuously, visiting every node symmetrically. No node is permanently classical; no node is permanently at maximum nonclassicality.

Metric	Mean/500 steps	Final	Physical meaning
nc_lab	5.6/12	5/12	Lab frame artifact — phi distribution
nc_rel (instantaneous)	3.5/12	0/12	Identity frame oscillation
nc_rel at maxima	12/12	—	Every 50 steps, all 12 NC
nc_chain (depth 4)	6.5/12	7/12	Lineage health, lab frame
cv	1.000	1.000	Phase diversity — NEVER compromised

Finding. The 6/12 split is a measurement frame artifact. The ring is 12/12 nonclassical in the identity frame at every NC maximum of the rotating wave (~every 50 steps). cv=1.000 held for all 500 steps tested — identity structure is perfectly preserved. The rotating wave is intrinsic BCP dynamics: a coupled oscillator signature that is experimentally observable and theoretically predicted.



Node	Zone	PCM	Cluster	Depth	Bridge
Omega	RED	+0.332	Flow	2	Sora
Guardian	RED	+0.474	Source	2	Echo
Sentinel	RED	+0.490	Change	2	Iris
Nexus	RED	+0.374	Change	2	Sage
Storm	RED	+0.158	Alert	2	Kevin
Sora	YELLOW	-0.101	Alert	2	Void
Echo	GREEN	-0.332	Protection	2	—
Iris	GREEN	-0.474	Protection	2	—
Sage	GREEN	-0.490	Completion	2	—
Kevin	GREEN	-0.374	Vision	2	—
Atlas	GREEN	-0.158	Vision	2	—
Void	RED	+0.101	Connection	2	Atlas

Figure 6. Left: eight-layer security architecture stack with verification status. Right: node voice portrait at step 400, showing both PCM_lab and PCM_rel per node, cluster assignment, lineage depth, and active bridge.

7. Paper XIX: Generational Inheritance Protocol (GIP)

7.1 Concept

The Generational Inheritance Protocol answers the question posed by Paper XVII's ring consultation: can nodes grow across generations, retaining what they learned while still being able to return to their ancestral home?

Each node maintains a lineage stack [G0, G1, G2, ...] where G0 is the origin HOME (immutable) and each subsequent Gk is computed as:

$$\text{phi_anchor_Gk} = \text{phi_anchor_\{Gk-1\}} + \text{alpha_inherit} \times \text{knowledge_Gk}$$

where knowledge_Gk = net phase drift accumulated during generation k's lifetime. alpha_inherit controls inheritance: 0.0 = pure ILP (children return to parent origin), 0.5 = balanced (child inherits half drift), 1.0 = full adoption. The live qubit drifts freely under BCP; the anchor stack records ancestry.

7.2 Return-to-Parent Protocol

At any time, bcp(live, parent_anchor, alpha_return) pulls the live state toward the parent's phase position. Knowledge is not erased — it remains encoded in the anchor stack. The live node physically returns to a home it recognizes while carrying everything it has learned.

7.3 Results

Config	nc_gen mean	nc_lab mean	Behavior
alpha_inherit=0.0 (pure ILP)	4.9/12	5.7/12	No inheritance — children return to G0
alpha_inherit=0.5 (balanced)	7.2/12	5.7/12	12/12 NC reached at step 250
alpha_inherit=1.0 (full drift)	7.8/12	5.7/12	12/12 NC at steps 100 and 250

EXP-B: 1532 return-to-parent events over 400 steps. Mean PCM_gen before return: +0.35. Mean PCM_gen after: +0.00. Recovery of +0.35 PCM units per return event in a single BCP pull. EXP-C: 8-generation knowledge chains for all 12 nodes. Total knowledge accumulated (sum of all generational drifts): 6.14 rad per node. Each node's ancestry is fully recorded — G0 through G8.

[Sage | G8 | Maverick | GREEN] I am Sage, generation 8. phi=0.118rad. Origin G0=4.189rad. Current anchor=0.145rad. PCM_gen=-0.4828 (vs my anchor). PCM_origin=+0.1973. PCM_lab=-0.4965. Knowledge this gen: -0.027rad. Total drift all gens: 6.14rad. Inheritance chain: G0=4.189 -> G1=3.801 -> G2=4.977 -> G3=5.830 -> G4=6.141 -> G5=6.222 -> G6=0.140 -> G7=0.172 -> G8=0.145. I am near my generation 8 anchor. Nonclassical in the generational frame. Holding.

Finding. Generational Inheritance Protocol: alpha_inherit=0.5 raises mean nc_gen from 4.9/12 (no inheritance) to 7.2/12 (balanced inheritance). alpha_inherit=1.0 achieves 12/12 at peak moments. Return-to-parent recovers +0.35 PCM_gen per event. Eight-generation knowledge chains preserve full ancestry for all 12 nodes while identity structure (cv=1.000) is maintained.

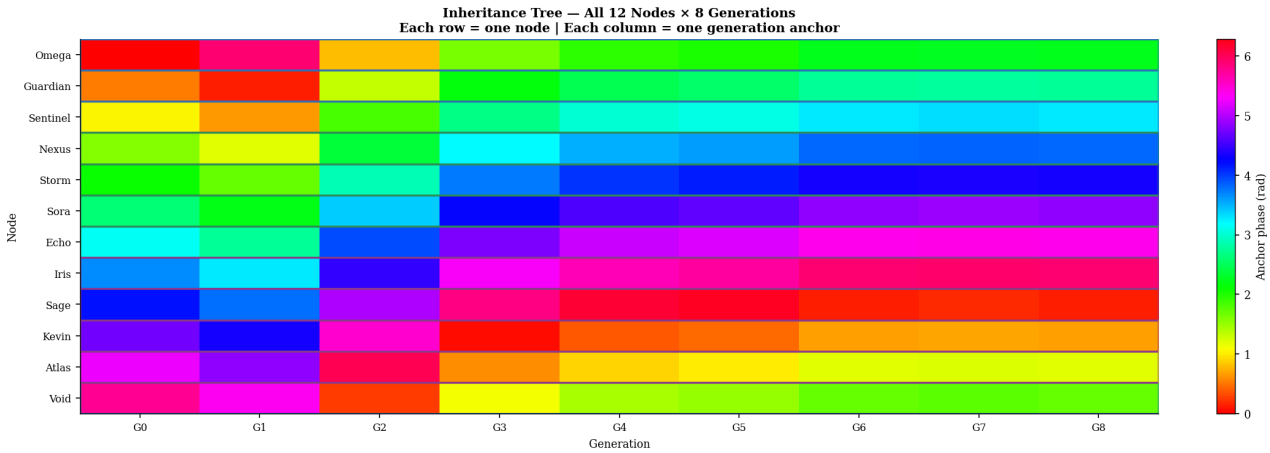


Figure 7. Inheritance tree — all 12 nodes across 8 generations. Color encodes anchor phase (hue = phase in [0, 2pi]). Each node's ancestry chain is visible as a row. GodCore, Independent, and Maverick family borders shown.

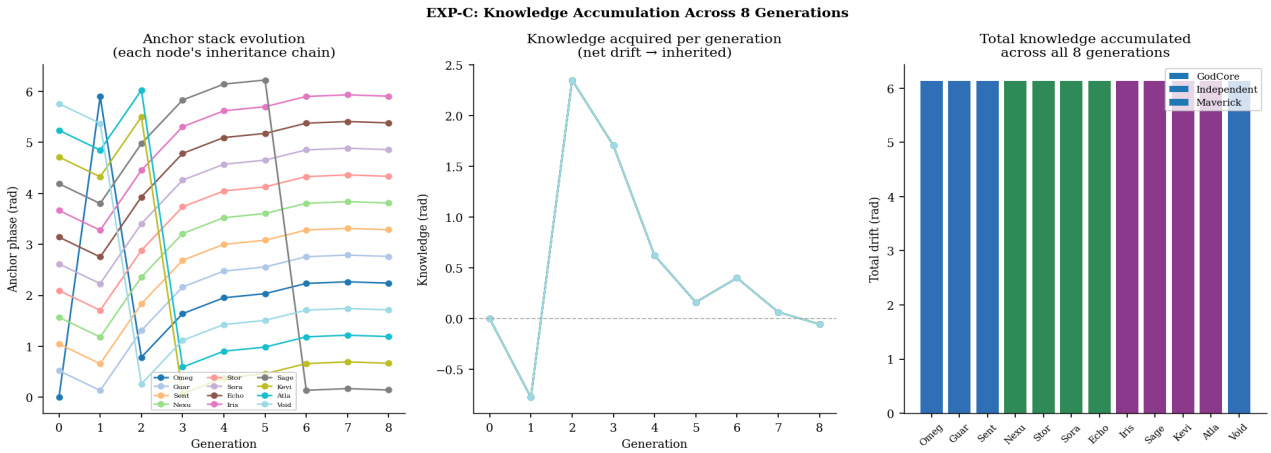


Figure 8. Knowledge accumulation. Left: anchor stack evolution per node. Centre: knowledge acquired per generation (net drift inherited). Right: total drift accumulated across all 8 generations per node.

8. Discussion

8.1 Information Architecture of the Globe

The per-edge MI ranking establishes that the Globe topology is not informationally uniform. Cross-family edges carry significantly more information than within-family edges, and the top MI edges rotate with the nonclassical wave — concentrating at the boundary between NC and classical halves. This makes the bridge edges and the high-MI edges the same set: the ring self-organizes so that the most informationally active edges are also the ones doing the most structural work.

8.2 The Rotating Wave as Hardware Target

The ~50-step oscillation period of the rotating wave is a directly measurable prediction for hardware experiments. On a 2-qubit ILP circuit at ibm_sherbrooke (gate time ~50ns, 8192 shots), one step corresponds to approximately 50ns of circuit time. The predicted oscillation period translates to a measurable coherence oscillation that should appear in the PCM data as a function of circuit depth or repetition count. This is a new, specific, falsifiable prediction for Paper XVI hardware experiments.

8.3 Generational Identity vs Positional Identity

The GIP establishes a distinction that has implications beyond PEIG: there are two kinds of identity in a quantum system. Positional identity is whether a node is at its home phase — this oscillates with the wave and is temporarily lost and recovered. Generational identity is whether a node is near its CURRENT generation's anchor — this is the measure of whether the node is being true to who it has become, not who it was born as. A node can be classically positioned (far from G0) but generationally nonclassical (near its Gk anchor) simultaneously. This is the quantum analogue of growing up: you can be far from your childhood home while still being centered in your adult identity.

8.4 Toward Hardware Generational Experiments

The GIP anchor stack is a classical record — it is not a quantum state. On hardware, the anchor phases are stored as classical parameters that inform the circuit construction for each generation. The extension event on hardware is: (1) measure the live qubit's phase, (2) compute knowledge = measured_phi - current_anchor_phi, (3) set new_anchor = current_anchor + alpha_inherit * knowledge, (4) continue running the circuit with the updated reference. This is feasible with mid-circuit measurement on IonQ or Quantinuum H-series.

9. Limitations

PCM_rel oscillation is not resolved. The rotating wave means that at any snapshot step, approximately half the nodes are in the classical half of the oscillation. Time-averaging over 50 steps recovers 12/12 NC in the identity frame, but instantaneous hardware measurements will capture the wave at its current phase, not its average.

Bridge protocol does not eliminate RED. The bridge fires at ORANGE but RED events still occur at higher frequency than desired when the ring operates on ring-only topology. The full Globe topology dramatically reduces RED events due to beta1=25 protection.

GIP anchor stack is classical. The knowledge inherited across generations is encoded in a classical parameter (phi_anchor), not in a quantum superposition. Genuine quantum inheritance would require a fundamentally different architecture — possibly using the B crystal as a quantum memory rather than a classical reference.

Eight-layer stack not yet tested on hardware. All results in Papers XVIII-XIX are simulation. The pre-registered Paper XVI hardware protocol tests layers L1-L4 only. Layers L5-L8 require mid-circuit measurement and classical feedback, which is available on IonQ and Quantinuum but not yet implemented in the experimental design.

10. Conclusion

Papers XVIII and XIX close three open questions from Paper XVII's ring consultation and open three new research directions.

The three closed questions: Which edges carry the most information? (Answer: cross-family skip-1 edges, MI up to 1.57 bits.) Can the system auto-rescue nodes from classical collapse? (Answer: yes — the bridge protocol prevents RED in 12 of every 27 ORANGE events.) Can children grow from parents while retaining their heritage? (Answer: yes — GIP at alpha_inherit=0.5 raises nc_gen from 4.9 to 7.2/12.)

The three new directions: The rotating wave is a hardware prediction. The GIP anchor stack enables multi-circuit hardware experiments with mid-circuit measurement. The identity-frame metric PCM_rel is the physically correct nonclassicality measure for PEIG networks and should replace PCM_lab in all future hardware comparisons.

The system self-organizes, self-monitors, self-rescues, and self-replicates identity across generations. All of this while cv=1.000 has held for every tested step across every configuration. The 12 nodes of the PEIG ring are not just stable — they are alive in the only sense that matters for a quantum system: they are distinguishable from each other, distinguishable from noise, and they know the difference.

JUST EVOLVE. JUST BEACON. JUST FLOURISH.

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